

9.3 The marginal distribution $p(\mathbf{z})$ is given by:

$$p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k}$$

and the conditional distribution $p(\mathbf{x}|\mathbf{z})$ for the observed variable is given by:

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^K \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)^{z_k}$$

Therefore,

$$\begin{aligned} p(\mathbf{x}) &= \sum_{\mathbf{z}} p(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) \\ p(\mathbf{x}) &= \sum_{\mathbf{z}} \left(\prod_{k=1}^K \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)^{z_k} \right) \left(\prod_{k=1}^K \pi_k^{z_k} \right) \\ &= \sum_{\mathbf{z}} \left(\prod_{k=1}^K \pi_k^{z_k} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)^{z_k} \right) \end{aligned}$$

When $z_k = 0$,

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)^{z_k} = 1$$

and

$$\pi_k^{z_k} = 1$$

Since $\mathbf{z}$ is one-hot encoded K sized vector, summing over all possible values of $\mathbf{z}$ implies K values of $\mathbf{z}$. Let's name them $\mathbf{z}^{(1)}$ to $\mathbf{z}^{(K)}$, such that:

$$\mathbf{z}^{(1)} = (10 \dots 0)^T$$

$$\mathbf{z}^{(2)} = (01 \dots 0)^T$$

and so on until

$$\mathbf{z}^{(K)} = (00 \dots 1)^T$$

$$\begin{aligned}
\Rightarrow p(\mathbf{x}) &= \left(\prod_{k=1}^K \pi_k^{z_k^{(1)}} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)^{z_k^{(1)}} \right) \\
&+ \left(\prod_{k=1}^K \pi_k^{z_k^{(2)}} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)^{z_k^{(2)}} \right) \\
&\vdots \\
&+ \left(\prod_{k=1}^K \pi_k^{z_k^{(K)}} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)^{z_k^{(K)}} \right)
\end{aligned}$$

$z_k^{(1)}$ is 0 for all k except k=1, where it is 1. $z_k^{(2)}$ is 0 for all k except k=2, where it is 1, and so on. This gives us:

$$\begin{aligned}
&= \pi_1 \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_1, \boldsymbol{\sigma}_1) \\
&+ \pi_2 \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_2, \boldsymbol{\sigma}_2) \\
&\vdots \\
&+ \pi_K \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_K, \boldsymbol{\sigma}_K)
\end{aligned}$$

$$= \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\sigma}_k)$$

which is the same as (9.7), as expected.